



Diesel Generator Set

mtu 12V4000 DS1750

380V – 11 kV/50 Hz/prime power/NOx emission optimized
12V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 1590 kVA - 1700 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 75% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NOx emission optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer	mtu		Total oil system capacity: l	260
Model	12V4000G14F		Engine jacket water capacity: l	160
Type	4-cycle		Intercooler coolant capacity: l	40
Arrangement	12V		Combustion air requirements	
Displacement: l	57.2			
Bore: mm	170			
Stroke: mm	210			
Compression ratio	16.4		Combustion air volume: m³/s	2.1
Rated speed: rpm	1500		Max. air intake restriction: mbar	50
Engine governor	ECU 9		Cooling/radiator system	
Max power: kWm	1420			
Air cleaner	dry			
Fuel system			Coolant flow rate (HT circuit): m³/hr	56
			Coolant flow rate (LT circuit): m³/hr	30
			Heat rejection to coolant: kW	590
			Heat radiated to charge air cooling: kW	360
Maximum fuel lift: m	5		Heat radiated to ambient: kW	75
Total fuel flow: l/min	16		Fan power for electr. radiator (40°C): kW	70
Exhaust system				
Fuel consumption ²⁾	l/hr	g/kwh	Exhaust gas temp. (after turbocharger): °C	470
At 100% of power rating:	373	218	Exhaust gas volume: m³/s	5.2
At 75% of power rating:	272	212	Maximum allowable back pressure: mbar	85
At 50% of power rating:	184.8	216	Minimum allowable back pressure: mbar	30

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NOx emission optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 S5 (Low voltage Leroy Somer standard)	380 V	1360	1700	2583	1288	1610	2446
	400 V	1360	1700	2454	1288	1610	2324
	415 V	1360	1700	2365	1288	1610	2240
Leroy Somer LSA52.3 S6 (Low voltage Leroy Somer oversized)	380 V	1360	1700	2583	1288	1610	2446
	400 V	1360	1700	2454	1288	1610	2324
	415 V	1360	1700	2365	1288	1610	2240
Marathon 743RSL7090 (Low voltage Marathon)	380 V	1352	1690	2568	1288	1610	2446
	400 V	1344	1680	2425	1288	1610	2324
	415 V	1272	1590	2212	1272	1590	2212
Marathon 744RSL7091 (Low voltage Marathon oversized)	380 V	1352	1690	2568	1288	1610	2446
	400V	1344	1680	2425	1288	1610	2324
	415V	1272	1590	2212	1272	1590	2212
Marathon 1020FDH7095 (Medium volt. marathon)	11 kV	1352	1690	89	1280	1600	84
Leroy Somer LSA53.2 VL6 (Med. volt. Leroy Somer)	11 kV	1352	1690	89	1288	1610	85

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.